

Sun God and Wind God team up to power-up India

"Solar energy and wind energy appear to be the most feasible energy technologies for India," comments Bobby Mohan, joint executive officer & head of business, Distil Information Systems. Recent news reports justify this point.

The Airvorce Group and Satluj Jal Vidyut Nigam, for example, have joined hands to build a 13GW green energy plant in Karnataka, wherein 10 GW will be from photovoltaic and 3 GW from wind farms.

This project is expected to outpace even China's wind energy development plans. Over \$50 billion will be invested over the next ten years to make the project a reality.

The 13GW project will be located in Karnataka, where twelve potential sites have been identified for the solar farms and four areas for the wind farms. The first phase of the development is likely to begin in the next three years and will have 100 MW of solar and 200 MW of wind power. Eurowinds and Suzlon are potential candidates for sourcing the turbines.



Wind farm

scale. A chunk of ceramic acts as an electrolyte, and allows ion transfer between the air and the fuel. This pushes power around an external circuit. The company says the electrolyte is painted with what it calls 'secret inks' on each side to act as anode and cathode.

Topsoe Fuel Cell in Denmark claims to have come up with a lower-cost model. The use of thick electrolytes in the Bloom fuel cell leads to high material costs as well as high operating temperatures—both significant cost points. Topsoe's model uses a thick anode to scale down the electrolyte thickness. This reduces the operating temperature from around 900°C to 750°C, thereby also reducing the cost.

The operating temperature is an important criterion as far as a fuel cell goes, because it needs an external heat supply to start up (thereafter, the exothermic reactions inside it keep it going). Hence reducing the operat-

ing temperature is a key innovation area for SOFCs. According to a recent report in the *New Scientist*, Ceres Power in the UK has come up with SOFCs that operate below 600°C, because these use an electrolyte that works at lower temperatures than those used by

Bloom and Topsoe. At such operating temperatures, the device can be held together using steel itself, eliminating the need for ceramics.

"A porous steel sheet at the cell's heart is printed with ultra-thin layers of ceramic anode, electrolyte and cathode. Home boilers powered by the cells are cheap enough to begin rolling them out in their thousands this year—the start of a four-year programme to install 37,500 in the homes of customers of the UK's biggest energy supplier, British Gas," states the report.

Fuel cells, in overdrive

Any mention of fuel cells immediately reminds us of companies such as Daimler, GM, Toyota and Ford, which have long been pursuing the dream of fuel-cell-powered cars. Without a doubt, those will be dream cars—emission-free, quiet, smooth and nature-friendly.

Fuel cell vehicles use polymer electrolyte membrane or proton exchange membrane (PEM) fuel cells, which directly convert the chemical energy bound in hydrogen into electrical energy that powers electric motors. These are subject to much lower energy loss than vehicles with an internal combustion engine and have twice their efficiency factor.

An individual fuel cell is only about two millimetres thick, but it generates only around 1 volt—certainly not enough to fire a car. Hence hundreds of cells are stacked to procure a few hundred volts to power a vehicle.

Despite their many advantages, fuel cell cars are still the privilege of a select few. As the cost of production of these cars is very high, they are still manufactured in 'prototype-scale' only. However, the quest to mass-produce fuel cells cars and other vehicles continues, and industry majors aim to have hydrogen fuel cell powered sedans in the market by 2015.

Tata Motors has developed a range of hydrogen fuel-cell-powered commercial vehicles like buses and light trucks, which are under trial in its European centre. Tata also has plans involving other alternative fuel technologies like bio-diesel, ethanol, hybrid, electricity and fuel cells to run the automobiles. You might recall that in 2006, the Tata Group had tied up with the Indian Space Research Organisation (ISRO) to design and run buses using hydrogen. It is also working on cars that run on hydrogen fuel cells.

Wind farms in deep waters—no puns intended!

Wind is a wonderful source of energy, abundant like solar and hydro power. But it needs to be managed and used well.

A dozen windmills put together as a farm can cause more noise than a class of kindergarten kids. Well, obviously, nobody would like to live next to a wind farm. One way out is to implement these wind farms 'off-shore,' that is, around 10 kilometres